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First paleomagnetic data for the New Siberian Islands: Implications for Arctic paleogeography

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ABSTRACT

In this paper we present new paleomagnetic and paleontological data from the Ordovician and Silurian carbonate rocks of Kotelny Island (the Anjou Archipelago), and from the Ordovician turbidities of Bennett Island (the De Long Archipelago). It is assumed that both archipelagos belong to the NSI (New Siberian Islands) terrane — a key tectonic element in the Arctic region. Ages of the studied rocks have been established by paleontological data and lithological correlations. Our new data on conodonts combined with those from previous studies of Ordovician and Silurian fauna indicate a biogeographic similarity between the shelves of the Siberian paleocontinent and the NSI in the Early Paleozoic. Three new paleomagnetic poles for the NSI (48.9°N, 13.8°E, A_{95} = 18.1° for 475 Ma; 45.5°N, 31.9°E, A_{95} = 11.0° for 465 Ma, and 33.7°N, 55.7°E, A_{95} = 11.0° for 435 Ma) fall between the south-eastern part of Central Europe and the Zagros Mountains. The similarity of paleomagnetic directions from Kotelny and Bennet islands confirms that both the Anjou and De Long archipelagos belong to the same terrane. Calculated paleolatitudes indicate that in Ordovician–Silurian times this terrane has been located between 30° and 45°, possibly in the northern hemisphere. Based on this observation, we suggest a linkage between the NSI and the Kolyma–Omolon superterrane. Comparison of apparent polar wander paths (APWPs) of the NSI, Siberia and other cratons/terranes suggests that the NSI drifted independently. We demonstrate that the structural line between Svyatoy Nos Peninsula and Great Lyakhovsky Island is the continuation of the Kolyma Loop suture on the Arctic shelf, and expect that the continuation of the South Anyui suture is to be found east of the NSI.

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1. Introduction

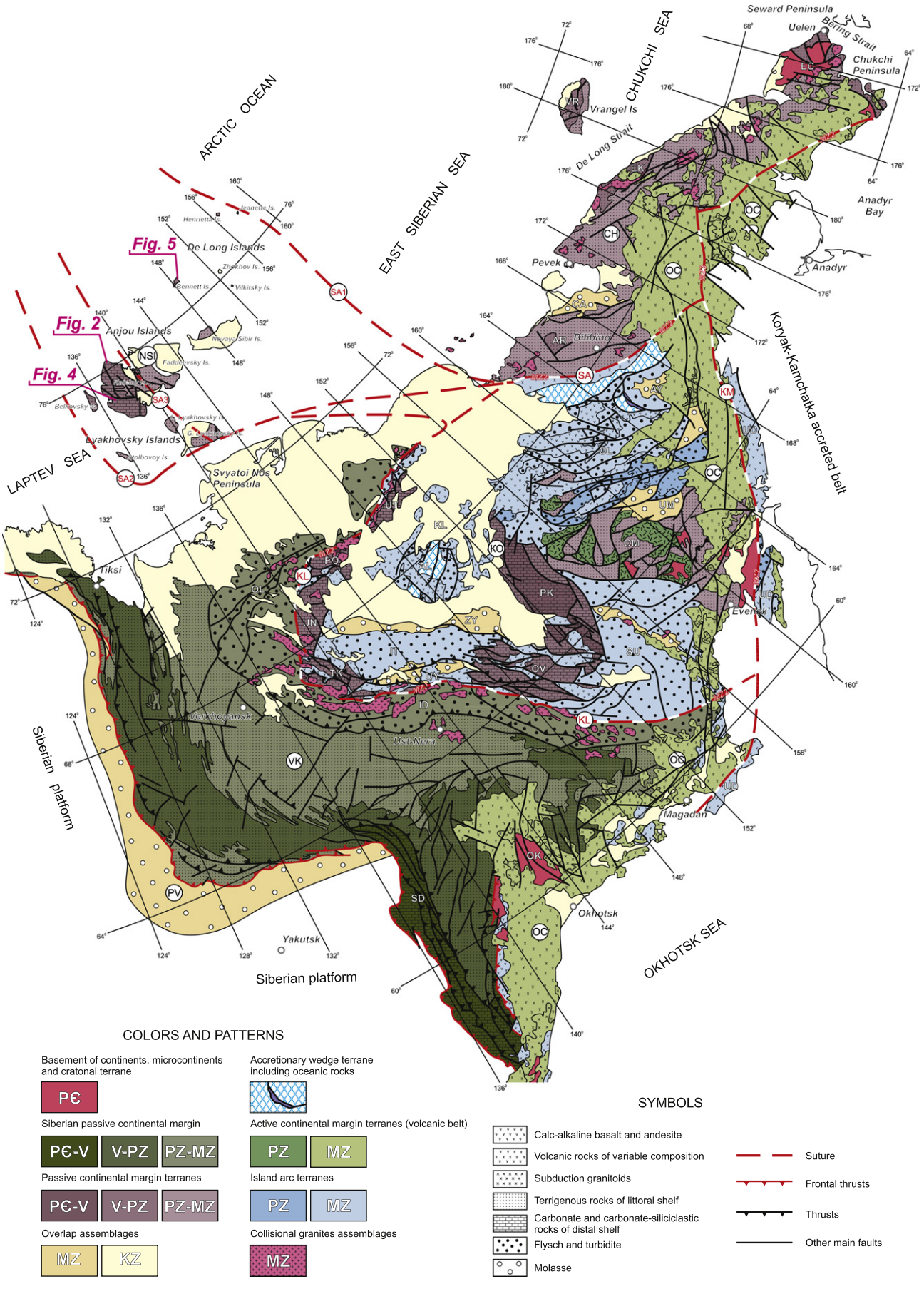
The NSI (New Siberian Islands) terrane is located on the Siberian Arctic shelf. The terrane includes the Lyakhovsky (Great Lyakhovsky, Little Lyakhovsky and Stolbovoy islands), Anjou (Kotelny, Faddeevsky, Belkovsky and Novaya Sibir islands) and De Long (Bennett, Vilkitsky, Zhokhov, Henrietta and Jannette islands) archipelagos (Fig. 1). The geological history of the NSI, its formation and initial affinity (Siberia vs Laurentia; Khain et al., 2009) are widely debated. For example, there is little consensus on the position of the western margin of the Alaska–Chukchi terrane, which may include the NSI (e.g. Fujita, 1978; Drachev et al., 1998; Drachev, 2002; Filatova and Khain, 2007; Kuzmichev, 2009). Therefore, the NSI terrane is a valuable target for paleomagnetic and geochronological studies, which may provide important clues for a better understanding of the main geological structures of NE Asia and their traces on the shelves of Laptev and East Siberian seas.

The presence of Precambrian metamorphic complexes within the main structures of the Arctic Ocean has led to an idea of a large continental mass between Laurentia, Baltica and Siberia in the past (Shatsky, 1963; Khain, 1979; Zonenshain and Natapov, 1987). This ancient continent is sometimes referred to as Hyperborea (Shatsky, 1963) or Arctida (Zonenshain and Natapov, 1987). The tectonic history of Arctida is not clear, but after its breakup various fragments have been probably accreted to larger continents of the Arctic realm (Zonenshain and Natapov, 1987; Lawver et al., 2002, 2011; Metelkin et al., 2015).

The final accretion of Arctida's fragments to other continents occurred in the Late Carboniferous–Early Permian (Zonenshain and Natapov, 1987) after the closure of the Paleo-Asian and Paleo-Urals oceans (Zonenshain et al., 1990; Dobretsov, 2003). At the same time

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ARCTIC OCEAN
EAST SIBERIAN SEA
CHUKCHI SEA
LAPTEV SEA
OKHOTSK SEA
Seward Peninsula
Uelen
Bering Strait
Chukchi Peninsula
Vrangel Is
De Long Strait
Anadyr Bay
Anadyr
Pevek
Koryak-Kamchatka accreted belt
Fig. 5
De Long Islands
Fig. 2
Anjou Islands
Fig. 4
Lyakhovsky Islands
Svyatoi Nos Peninsula
Tiksi
Verkhoyansk
Ust-Nera
Magadan
Okhotsk
Yakutsk
Siberian platform
Siberian platform
COLORS AND PATTERNS
Basement of continents, microcontinents and cratonal terrane
Pϵ
Siberian passive continental margin
Pϵ-V
V-PZ
PZ-MZ
Passive continental margin terranes
Pϵ-V
V-PZ
PZ-MZ
Overlap assemblages
MZ
KZ
Accretionary wedge terrane including oceanic rocks
Active continental margin terranes (volcanic belt)
PZ
MZ
Island arc terranes
PZ
MZ
Collisional granites assemblages
MZ
SYMBOLS
Calc-alkaline basalt and andesite
Volcanic rocks of variable composition
Subduction granitoids
Terrigenous rocks of littoral shelf
Carbonate and carbonate-siliciclastic rocks of distal shelf
Flysch and turbidite
Molasse
Suture
Frontal thrusts
Thrusts
Other main faults

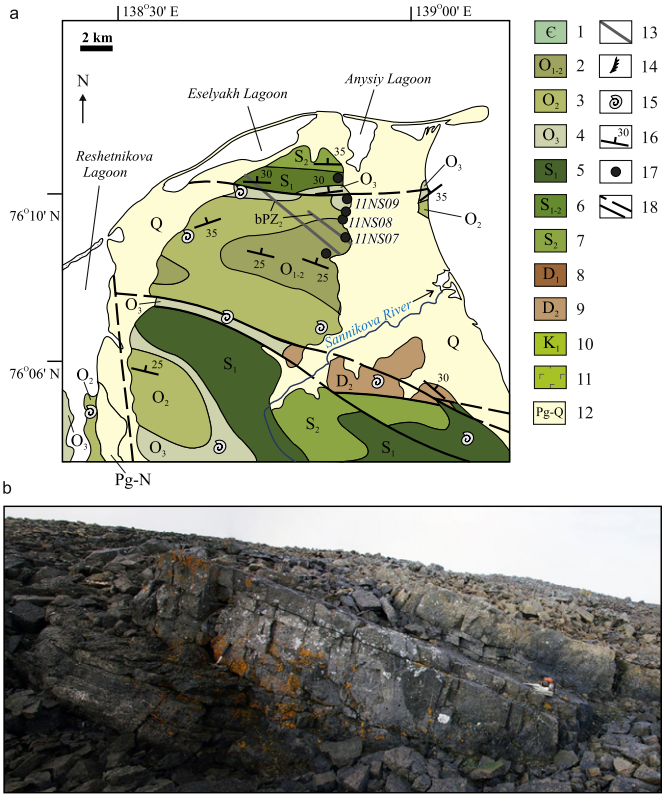


Fig. 2. Geological map of the Anisiy lagoon area (a) and a photograph of sampling site 11ns08 (b). Sedimentary rocks (1–10): 1 — Cambrian; 2–4 — Ordovician; 5–7 — Silurian; 8–9 — Devonian; 10 — Cretaceous (see Fig. 5); 11 — Cretaceous basaltic covers of Bennett Island (see Fig. 5); 12 — Paleogene–Quaternary cover; 13 — Late Paleozoic mafic dikes; 14–15 — fauna findings; 16 — bedding; 17 — paleomagnetic sampling sites; and 18 — faults.

the South Anyui (aka Proto-Arctic) Ocean has been formed as a bay of the Paleo-Pacific Ocean between Siberia and Laurentia (Zonenshain et al., 1990). Consequently some fragments of Arctida have been displaced during the Jurassic–Cretaceous rifting events (Grantz et al., 1998; Bogdanov, 2004; Shipilov, 2004; Khain et al., 2009; Drachev et al., 2010; Dobretsov et al., 2013; Laverov et al., 2013). In particular, the large Alaska–Chukchi fragment broke away from northern Laurentia during the opening of the Amerasian basin between 136 and 100 Ma (Grantz et al., 1998; Shipilov, 2004). The South Anyui Ocean simultaneously was closing on the opposite side of the Alaska–Chukchi terrane. This closure caused the formation of an elongated ophiolite zone — the South Anyui suture (Sokolov et al., 2001; Oxman, 2003; Sokolov et al., 2010). Prior to these events, the Kolyma–Omolon superterrane has been amalgamated by collision of different blocks including fragments of passive continental margins and island arcs. Consequently this superterrane collided with the Verkhoyansk margin of Siberia, forming the Kolyma Loop (Polousny–Kolyma) suture (Parfenov et al., 2003). Paleomagnetic studies in the western part of the Omolon massif (Bondarenko and Didenko, 1997; Didenko et al., 2002) indicate significant lateral displacements of the Kolyma–Omolon superterrane during that collision.

The continuation of the South Anyui suture on the modern Arctic shelf is a subject of recent debates (Franke et al., 2004, 2008; Kuzmichev, 2009). There are three alternative hypotheses. The first one suggests that the Anyui suture (SA in Fig. 1) continues to the shelf approximately along the 158th meridian east of the NSI (SA1 in Fig. 1) (Rowley and Lottes, 1988). According to the second hypothesis the suture follows to the shelf towards Great Lyakhovsky Island and then turns to the north just SW of Stolbovoy Island (Fig. 1) bounding the NSI terrane in the west (SA2 in Fig. 1) (Kuzmichev, 2009). The third hypothesis considers that the shelf continuation of the Anyui suture cuts through the NSI dividing the terrane into western ("Siberian") and eastern ("Chukchian") parts (SA3 in Fig. 1) (Spektor et al., 1981; Parfenov et al., 1993).

Here we present new geological and geophysical data to test these hypotheses. We also discuss main tectonic features of the NSI and their Early Paleozoic paleogeography.

2. Geology of the study area

2.1. General information

The New Siberian Islands (NSI) terrane is located between the Laptev and East Siberian seas in the eastern Arctic (Fig. 1). Traditionally, the Lyakhovsky and Anjou archipelagos are referred to as parts of a Middle Cretaceous orogen, which is mainly composed of the deformed passive continental margin of the Alaska–Chukchi terrane (Khain et al., 2009). Outcrops of pillow basalts have been reported from the southern part of Great Lyakhovsky Island (Shalaurova Peninsula) (Drachev and Savostin, 1993). These pillow basalts were usually considered to be relicts of the South Anyui Ocean (Kuzmichev, 2009). Their K–Ar and ^{40}Ar–^{39}Ar ages are between 170 and 160 Ma (Middle Jurassic; Kuzmichev and Lebedev, 2008; Kuzmichev, 2009). Other traces of the Late Mesozoic suture have been found in the northern part of the Svyatoi Nos Peninsula (Fig. 1). Here the Upper Jurassic basalts and andesites are suggested to be associated with a wide accretionary wedge of the Anyui–Svyatoi Nos paleoisland arc (Kuzmichev et al., 2005; Kuzmichev, 2009).

Most researchers consider a post-Grenvillian age for the NSI basement as well as for the entire basement of the Alaska–Chukchi terrane (Zonenshain and Natapov, 1987; Bogdanov, 2004; Kos'ko and Korago, 2009). However, this is not yet confirmed by any reliable geochronological data. Recently Korago et al. (2014) reported U–Pb SHRIMP/SIMS ages of 661–637 Ma on monzodiorite porphyrites, dolerites and tuffaceous-sedimentary rocks of Henrietta Island. The main phase of deformations in the NSI is probably related to the Jurassic–Cretaceous collision event after the closure of the South Anyui Ocean (e.g. Brandes et al., 2015). Some indications of the Caledonian orogeny north of the De Long Islands have been reported by Kos'ko (1977) and by Filatova and Khain (2007). For example, Filatova and Khain (2007) described the Ordovician (c. 440 Ma) turbidites, basalts, andesite-basalts, and associated intrusions of diorites and dolerites in Henrietta Island. The volcano-sedimentary complex of the nearby Jeannette Island (Fig. 1) has a similar structure, and probably a similar age, as suggested by preliminary radiometric ages of some dolerite dykes exposed in this island (Sobolev et al., 2014). Filatova and Khain (2007) interpreted these rocks of Henrietta and Jeannette islands as parts of a suprasubduction complex. These rocks may have experienced some deformations during the 364–342 Ma Ellesmerian orogeny, as indicated by a structural unconformity observed on the Anjou Archipelago (Filatova and Khain, 2007).

The isometric circular De Long structure is expressed as a single volcanic dome, which is separated from the Anjou Archipelago by the

Fig. 1. Geological map of northeast Russia (after Tilman and Bogdanov, 1992; Parfenov and Kuzmin, 2001). Main tectonic units (in circles): PV — Priverkhoyansk foreland basin; VK — Verkhoyansk fold belt, including Inyali–Debin (ID) and Oldjoy (OL) turbidite terranes, Okhotsk (OK) metamorphic terrane and Sette–Daban (SD) uplift; KO — Kolyma–Omolon superterrane, including Omolon microcontinent (OM); passive continental margin terranes: Prikolyma (PK) and components of Omulevka: Omulevsky (OV), Tas–Khayakhtas (TK), Uyandina (UN), Polousny (PO), and Ulakhan–Tas (UT); accretionary prism and island arc terranes: Sugoy (SU), Alazey (AL), Oloy (OL), Uyandino–Yasachnen (UY), Ilin'–Tass (IT) and overlying sedimentary basins: Zyrian (ZY), Umkuveem (UM), Aynakhurgen (AY), and Kolyma (KL); Okhotsk–Chukotka volcanic belt (OC), including underlying fragments of the Uda–Murgal island arc (UD); Chukchi folded area (CH) including uplifts: Vrangel (VR), East-Chukchi (EC), Ekiatap (EK), Alyarmaut (AR) and the Chaun (CA) overlying sedimentary basin; NSI — New Siberian Islands uplift. Sutures: KL — Kolyma Loop (Kolyma–Polousny), SA — South Anyui (SA1–SA3: possible continuations), KM — Koni–Murgal.

modern trough formed over the New Siberian graben (Drachev, 2002). Based on seismic data, this structure as well as the Vil'kitsky graben, that frames the De Long dome on the other side, has been formed in the Cretaceous (Drachev et al., 2001). Igneous rocks of the De Long Archipelago are also Cretaceous or slightly younger (Bogdanovsky et al., 1992; Fedorov et al., 2005). According to the widely accepted rotational model of the opening of the Amerasian basin (e.g. Rowley and Lottes, 1988; Grantz et al., 1998; Lawver et al., 2002), the margin between the Anjou and De Long archipelagos should represent a main suture between Siberia and the Alaska–Chukchi terrane.

This study was focused on the Paleozoic sedimentary complex of the NSI. This complex includes typical shelf deposits of a marginal sea (Kos'ko, 1977; Parfenov and Kuzmin, 2001; Drachev, 2002). The relatively shallow-water carbonate rocks are typical for the Anjou Archipelago. In contrast, deep-water clastic rocks are common in the De Long Archipelago. We studied paleomagnetically and paleontologically two sedimentary sections — one in Kotelny Island (the most complete section) and the other in Bennett Island.

2.2. Kotelny Island

The Ordovician to Middle Devonian sedimentary rocks of Kotelny Island are characterized by two major lithologies. Shallow-water limestones and dolomites are common in the northeastern part of the island. The thickness of the strata decreases to the southwest, where the deeper marine carbonates are interbedded with the fine grained clastic rock (Kos'ko, 1977). This is indicative for a littoral shelf sedimentary environment on a passive continental margin. The carbonates are unconformably overlain by the uniform Upper Devonian–Lower Carboniferous terrigeneous strata (Kos'ko, 1977; Filatova and Khain, 2007). Our studies were focused on the Ordovician–Silurian interval of the section in northeastern and southwestern parts of the island.

In the northeastern part of the island we sampled the lowermost parts of the section, represented by the Lower Ordovician Ayan Formation that crops out close to the Anisiy Lagoon (Fig. 2). This formation mainly consists of light-colored dolomites interbedded with dark-gray fine-grained sandstones and siltstones with carbonate matrix, and with few layers of detrital limestones. This sequence was studied in several small outcrops in the core of a low angle anticline along the bank of the Grebenochnaya River (sites 11ns07 and 11ns08 in Fig. 2). A previous geological study of this area (Kos'ko, 1977) revealed the presence of angarellas fossils (*Angarella* cf. *lopatini Asatkin*), trilobites, rare brachiopods, and ostracods. These fauna fossils suggest the Late Floian (477–470 Ma) age of the Ayan Formation. A conodont complex containing *Semiacontiodus* sp. and *Drepanodus* sp. (Fig. 3) has been identified during our study. Observed fossils imply a slightly older age of 480–475 Ma for the sampled dolomites. When interpreting paleomagnetic data we used the upper age limit of 475 Ma.

In the southwestern part of the island we studied an extensive sequence of the Ordovician–Silurian rocks along the Chokurka River (Fig. 4). These rocks form two anticlines. Middle to Upper Ordovician Malodiring–Ayan and Terutekh formations are accessible in the cores of these anticlines. Lower Silurian Urasin and Nikol'skaya formations, in turn, constitute their limbs. For the paleomagnetic analysis we sampled the upper part of the Terutekh Formation in two outcrops (sites 11ns01 and 11ns02 in Fig. 4) and two sites from the basal layers

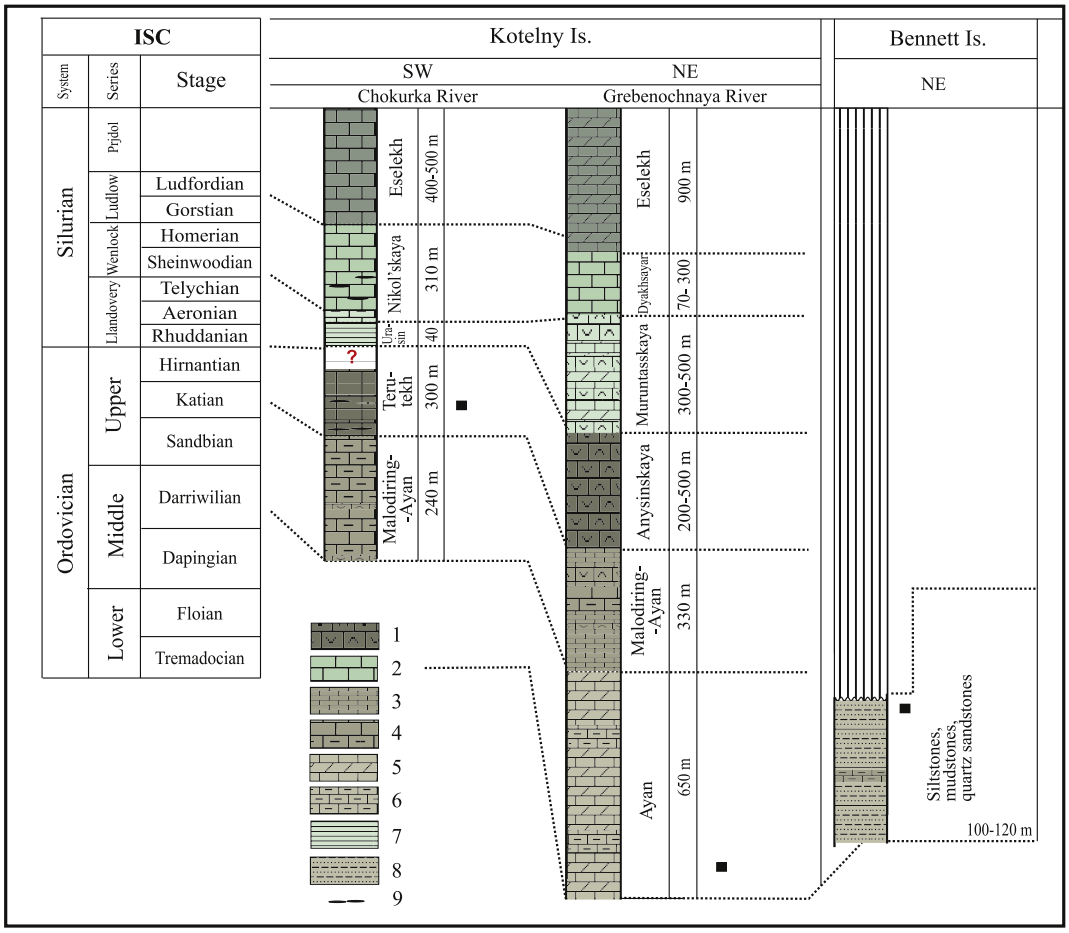


Fig. 3. Generalized stratigraphic chart of the Ordovician and Lower Silurian strata in the New Siberian Islands. 1 — bioclastic limestones; 2 — mostly micritic limestones; 3 — limestones and siltstones interlayering; 4 — limestones and marls interlayering; 5 — dolomites; 6 — marls, silty limestones; 7 — mudstones, siltstones; 8 — sandstones, siltstones; and 9 — chert lenses. Squares indicate paleomagnetic sampling.

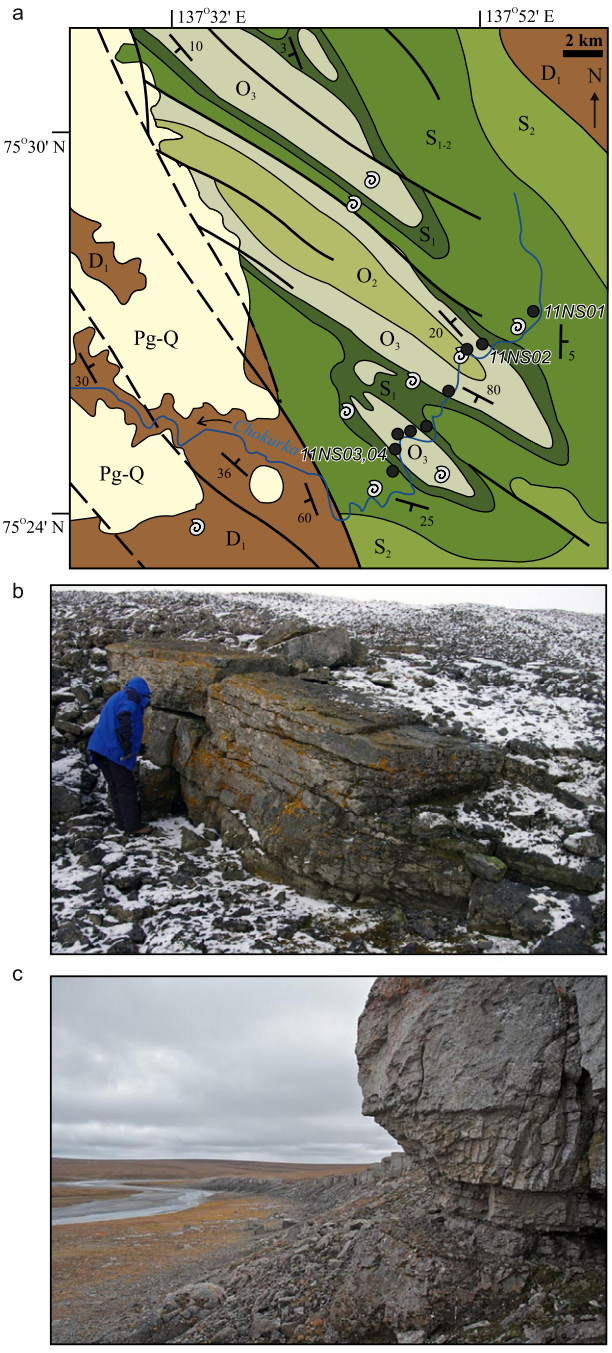


Fig. 4. (a) Geological map of the Chokurka River area; (b–c) photographs: sampling site 11ns01, Terutekh Formation (b) and sampling site11ns03, Urasin Formation (c). Legend — see Fig. 2.

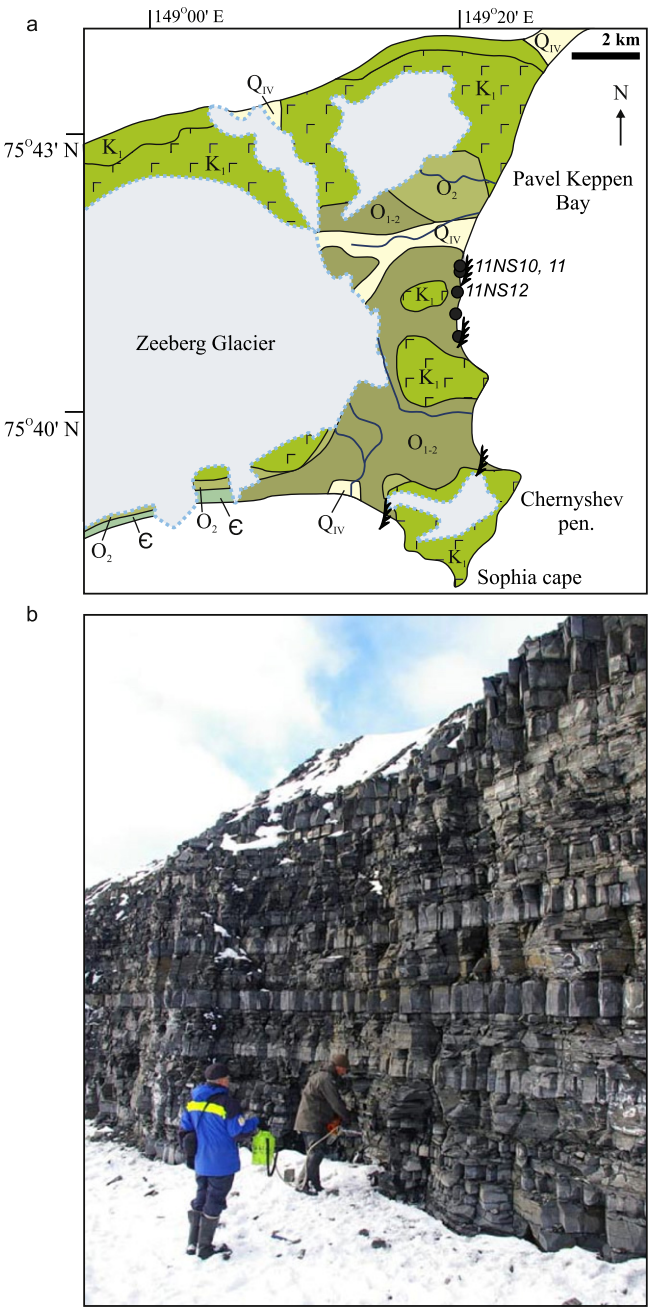


Fig. 5. Geological map of the north-eastern coast of Bennett Island (a) and the photograph of sampling site 11ns10 (b). Legend — see Fig. 2.

of the Urasin Formation (sites 11ns03 and 11ns04 in Fig. 4). The sampled part of the Terutekh Formation is composed of massive fine-grained, sometimes argillaceous limestones interbedded with fossil-rich limestones containing brachiopods, corals, gastropods and trilobites. An abundant tabulata complex indicates the age of the formation between Katian and Hirnantian stages (453 and 443 Ma) (Kos'ko, 1977). These rocks correlate with the Nalchanskaya Formation in the Omulevka Mountains of the Kolyma–Omolon superterrane, and with Dolborian and Burian horizons on the eastern margin of Siberia.

A variety of conodonts has been found in the Terutekh Formation. These fossils include *Yaoxianognathus*? *tunguskaensis* (Moskalenko), *Paroistodus*? cf. *nowlani* Zhen, *Walliserodus* sp., *Belodina* cf. *B. compressa*

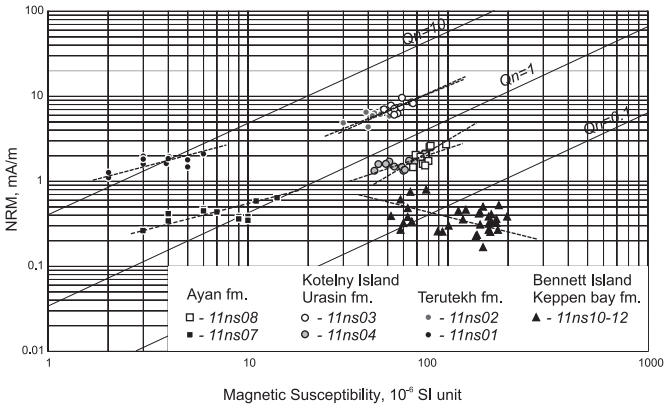


Fig. 6. Natural remanence magnetization (NRM) vs magnetic susceptibility (χ) diagram.

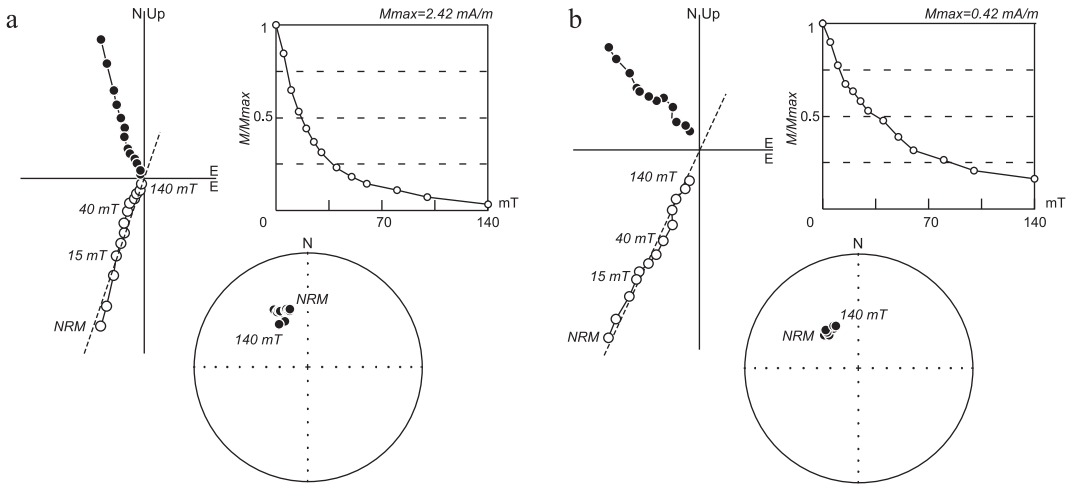


Fig. 7. AF demagnetizations of Ayan dolomites (Lower Ordovician, Kotelny Island). Sites: (a) 11ns08 (sample 11ns112) and (b) 11ns07 (sample 11ns093). In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots (equal-area projection) show upwards (downwards) pointing paleomagnetic directions with open (closed) symbols.

(Branson and Mehl), *Panderodus* ex gr. *greenlandensis* (Armstrong), *Pseudobelodina*? cf. *repens* (Moskalenko) and *Pseudobelodina* sp. (Fig. 3) and suggest the Upper Katian age (453–445 Ma) of the sampled rocks. Only some species vary within the formation. Based on these paleontological data, we interpret the age of sampled units as c. 450–444 Ma.

The Urasin Formation conformably overlies the Terutekh Formation and visually differs from the latter by a darker color (sometimes black). The studied outcrops are composed of thinly tabulated limestones, interbedded with calcareous black mudstones. The studied graptolites, brachiopods and rare trilobites (Kos'ko, 1977) suggest the Llandovery age (443–436 Ma) for the Urasin Formation. We use an age of 440 Ma in our interpretation of paleomagnetic data.

2.3. Bennett Island

The Lower Paleozoic section in Bennett Island is represented mostly by turbidites. Three sampling sites are located within an extensive coastal outcrop in the southern part of Pavel Keppen Bay (Fig. 5). The exposure contains cyclic shallow-dipping sequence of interbedded sandstones, siltstones, and mudstones (sites 11ns10, 11ns11 and 11ns12 in Fig. 5). Massive fine grained gray argillaceous sandstones from the bottoms of the Bouma cycles were most suitable for paleomagnetic analyses. Within most cycles they are overlain by cross-bedded fine-grained sandstones and siltstones. Thin-bedded platy mudstones with clearly defined syn-depositional slump structures usually form the top of the cycle. In general, separate cycles are characterized by decreasing thicknesses and sometimes by exclusion of their lower parts. For instance there are few cases of a singular interbedding of coarse and medium grained sandstones with graded bedding. Such features are typical for distal turbidites in a relatively close proximity to the provenance area.

Multiple Early and Middle Ordovician graptolites are widespread in the turbidites (Sobolevskaya, 1976). Our new paleontological observations of graptolites (Fig. 3) confirm this age. In the new collection of graptolites Sobolevskaya has identified: *Pseudogoniagraptus timidus*

Table 1
Paleomagnetic data from Early Paleozoic sedimentary rocks of the New Siberian Islands.

Sample site, rock, formation	*n/N*	In situ		Tilt corrected		*k*	α_{95}	*VGPole*			PL
		D (°)	*I* (°)	*D* (°)	*I* (°)			*PLat*	*PLong*	*dp/dm*	
Lower Ordovician, Kotelny Island											
11ns08, dolomite, Ayan fm.	11/12	273.1	66.3	320.7	58.2	24.6	9.4	48.9	7.5	13.9/10.3	38.9 ± 11.9
11ns07, dolomite, Ayan fm.	12/13	335.1	87.5	310.2	59.8	82.7	4.8	48.6	20.0	7.2/5.5	40.7 ± 6.3
Mean for 475 Ma dolomite of Ayan fm.		278.3	77.5			25.7	51.4				
				315.6	59.1	414.0	12.3	48.9	13.8	$A_{95} = 18.1$	39.9 ± 15.9
Middle Ordovician, Bennett Island											
11ns10, sandstone, Keppen bay fm.	8/11	299.8	52.3	307.8	56.3	27.7	10.7	44.2	31.0	15.4/11.1	36.9 ± 13.1
11ns11, sandstone, Keppen bay fm.	9/11	301.8	47.8	312.4	52.6	40.4	8.2	41.5	24.8	11.3/7.8	33.2 ± 9.4
11ns12, sandstone, Keppen bay fm.	8[a]/11	292.0	61.9	301.4	63.1	11.3	17.2	50.2	41.1	28.2/22.2	44.6 ± 25.0
Mean for 465 Ma sandstone of Keppen bay fm.		298.5	54.1			109.3	11.9				
				307.7	57.4	178.4	9.3	45.5	31.9	$A_{95} = 11.0$	38.0 ± 11.6
Upper Ordovician — Llandovery, Kotelny Island											
11ns02, limestone, Terutekh fm.	12/12	122.2	87.9	278.7	52.0	302.2	2.5	33.6	48.7	3.4/2.3	32.6 ± 2.8
11ns01, limestone, Terutekh fm.	10/11	231.8	71.3	269.7	53.9	102.2	4.8	33.1	53.9	6.7/4.7	34.4 ± 5.6
11ns03, limestone, Urasin fm.	11/11	182.9	83.1	272.8	55.8	87.8	4.9	35.7	55.6	7.0/5.0	36.3 ± 5.9
11ns04, limestone, Urasin fm.	10/11	195.9	79.3	266.8	53.4	87.2	5.2	31.9	60.4	7.2/5.0	34.0 ± 6.0
Mean for 435 Ma limestone of Terutekh and Urasin fm.		208.3	81.5			87.0	9.9				
				272.0	53.9	558.5	3.9	33.7	55.7	$A_{95} = 5.1$	34.4 ± 4.6

Note: *n/N* — number of samples used in statistics/total number of studied samples; *D* — paleomagnetic declination; *I* — paleomagnetic inclination; *k* — precision parameter; α_{95} — radius of the 95% confidence limit; *VGPole* — virtual geomagnetic pole: *PLat* — latitude (N degrees), *PLong* — longitude (E degrees), *dp/dm* — semiaxes of 95% confidence about the pole; the mean (paleomagnetic) pole was calculated as the mean of *VGPole*, where A_{95} — radius of the 95% confidence ellipse for the pole; PL — paleolatitudal position for the NSI (N degrees).

[a] Include four determinations by the great circles method.

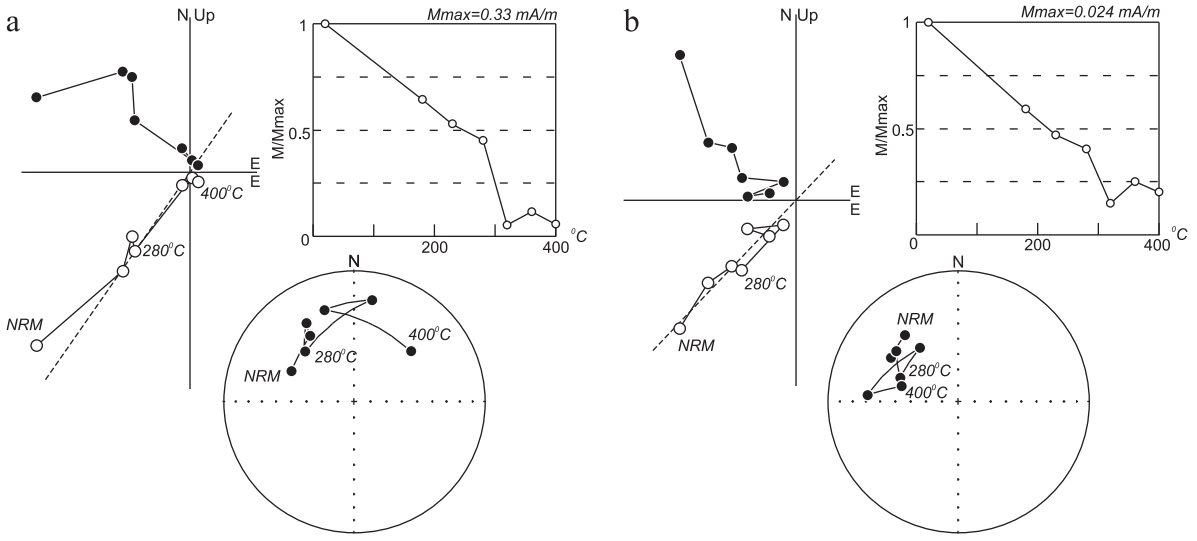


Fig. 8. Thermal demagnetizations of Keppen Bay sandstones (Middle Ordovician, Bennett Island). (a) Site 11ns10 (sample 11ns131); (b) site 11ns11 (sample 11nns141). See notes to Fig. 7.

(Harris et Thomas), *Expansograptus* sp., *Cryptograptus schaeferi* Lapworth, *Glossograptus* sp., *Paraglossograptus tentaculatus* (J. Hall), *Diplograptus* sp., *Climacograpthus* sp., *Eoglyptograptus* aff. *dentatus* (Brongniart), *Cryptograptus tricornis* (J. Hall), *Phyllograptus* aff. *anna* (J. Hall), and *Glossograptus* aff. *hincksii* (Hopkinson). All these taxons are widely distributed geographically and are typical for deep open sea basins at the beginning of the Darriwilian Age that corresponds to the *Expansograptus hirundo* and *Dydimograptus murchisoni* zones of the graptolite scale between 468 and 465 Ma. For paleomagnetic interpretation we used the youngest age of 465 Ma.

3. Paleomagnetic data

Paleomagnetic analyses have been carried out in the Paleomagnetic Center of the Institute of Petroleum Geology and Geophysics (Siberian branch of the Russian Academy of Science). Magnetic remanence was measured with the 2G Enterprises (USA) cryogenic magnetometer and the JR-6 (Czech Republic) spinner magnetometer. The sandstones from Bennett Island were subjected to a stepwise thermal demagnetization up to 600 °C using a MMTD80 thermal demagnetizer made by Magnetic Measurements Ltd., UK. Stepwise alternating field (AF) demagnetization of the Kotelny Island carbonates was performed using an automatic degaussing system built into the 2G Enterprises magnetometer.

Changes of the intensity and directions of remanent magnetization vectors during demagnetization experiments were analyzed using orthogonal vector projections (Zijderveld, 1967). The directions of natural remanent magnetization (NRM) components were identified using Principal Component Analysis (PCA) (Kirschvink, 1980) or sometimes remagnetization great-circle method (Halls, 1976). Site-mean and group-mean paleomagnetic directions were calculated using Fisher (1953) statistics or McFadden and McElhinny's (1988) statistics when PCA data were combined with those of the remagnetization circles analysis. The data were processed using software developed by Enkin (1994).

3.1. Kotelny Island, Lower Ordovician Ayan Formation

The NRM of dolomites varies from 1.5–3.0 mA/m in the upper part of the section (site 11ns08) to 0.1–0.3 mA/m in the lower part of the

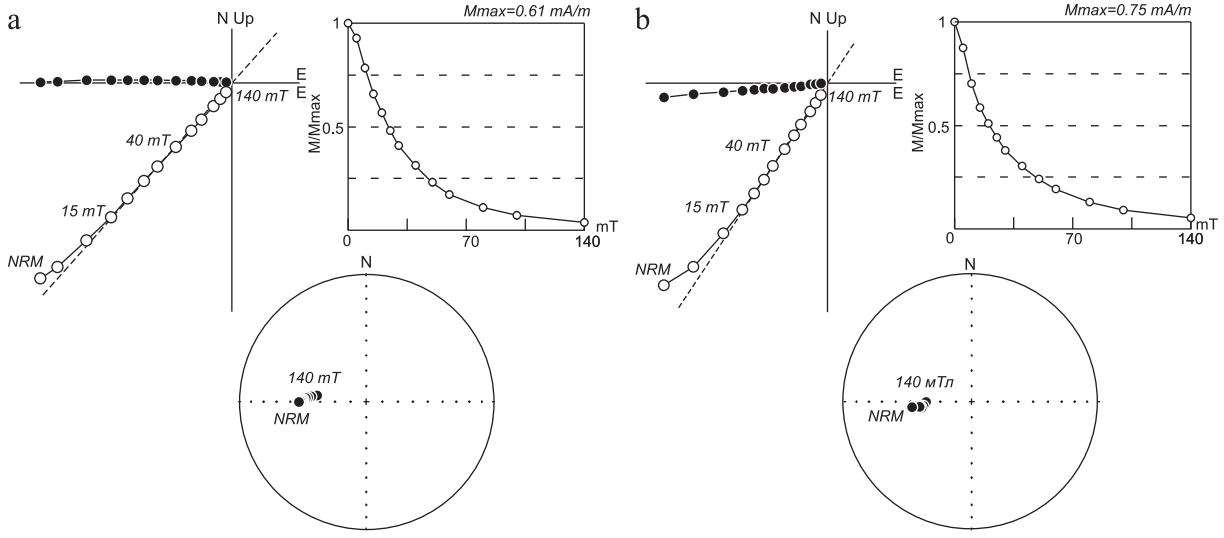


Fig. 9. AF demagnetization Upper Ordovician–Silurian limestones (Kotelny Island). (a) Site 11ns02 (Terutekh Formation, sample 11ns02); (b) site 11ns03 (Urasin Formation, sample 11ns023). See notes to Fig. 7.

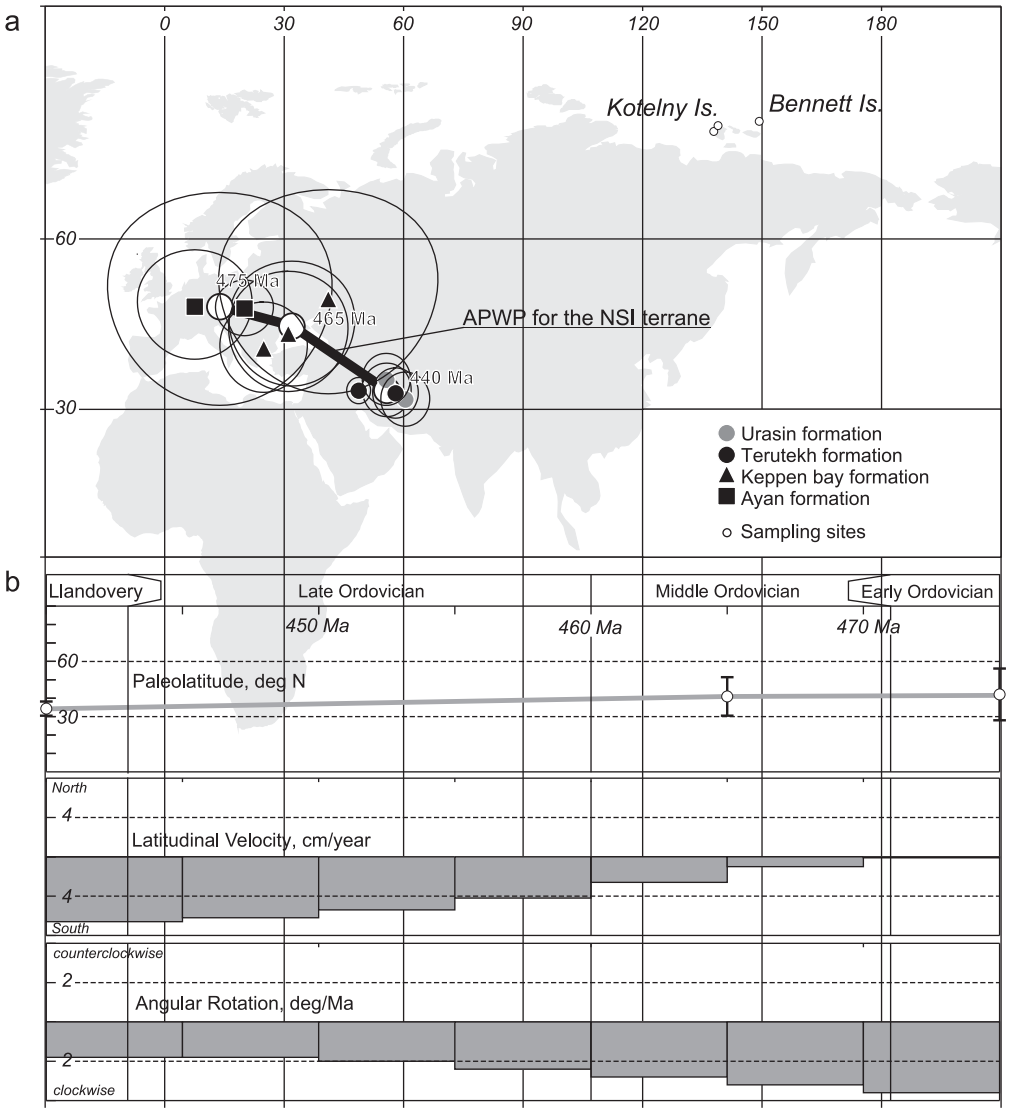


Fig. 10. (a) Virtual geomagnetic poles for Kotelny and Bennett islands (Table 1), mean paleomagnetic poles (open circles, Table 2) and the Ordovician–Silurian APWP of the NSI; (b) kinematic parameters.

section (site 11ns07). The magnetic susceptibility (χ) varies from ~10^{-4} SI units (upper part) to ~10^{-5} SI units (lower part). The Koenigsberger ratio Q is close to 1 in both parts (Fig. 6), which is typical for detrital remanent magnetization (DRM) (Nagata, 1965).

Both high- and low-coercivity components of NRM have been isolated by AF demagnetization (Fig. 7a). However, only a high-coercivity component has been found in some samples (Fig. 7b). The directions of components are close (Fig. 7a).

The median destructive field (MDF) for site 11ns08 (upper part of the section) is ~15 mT, indicating that the contribution of low-coercivity ferromagnetic minerals such as multi-domain magnetite into the NRM is significant. However, most of this contribution has been removed after 5–50 mT and the high-coercivity characteristic remanent magnetization (ChRM) has been sometimes destroyed only at 140 mT (Fig. 7a). Such demagnetization behavior suggests that the single-domain (SD) or pseudo-single-domain (PSD) magnetite might be the major ChRM carrier.

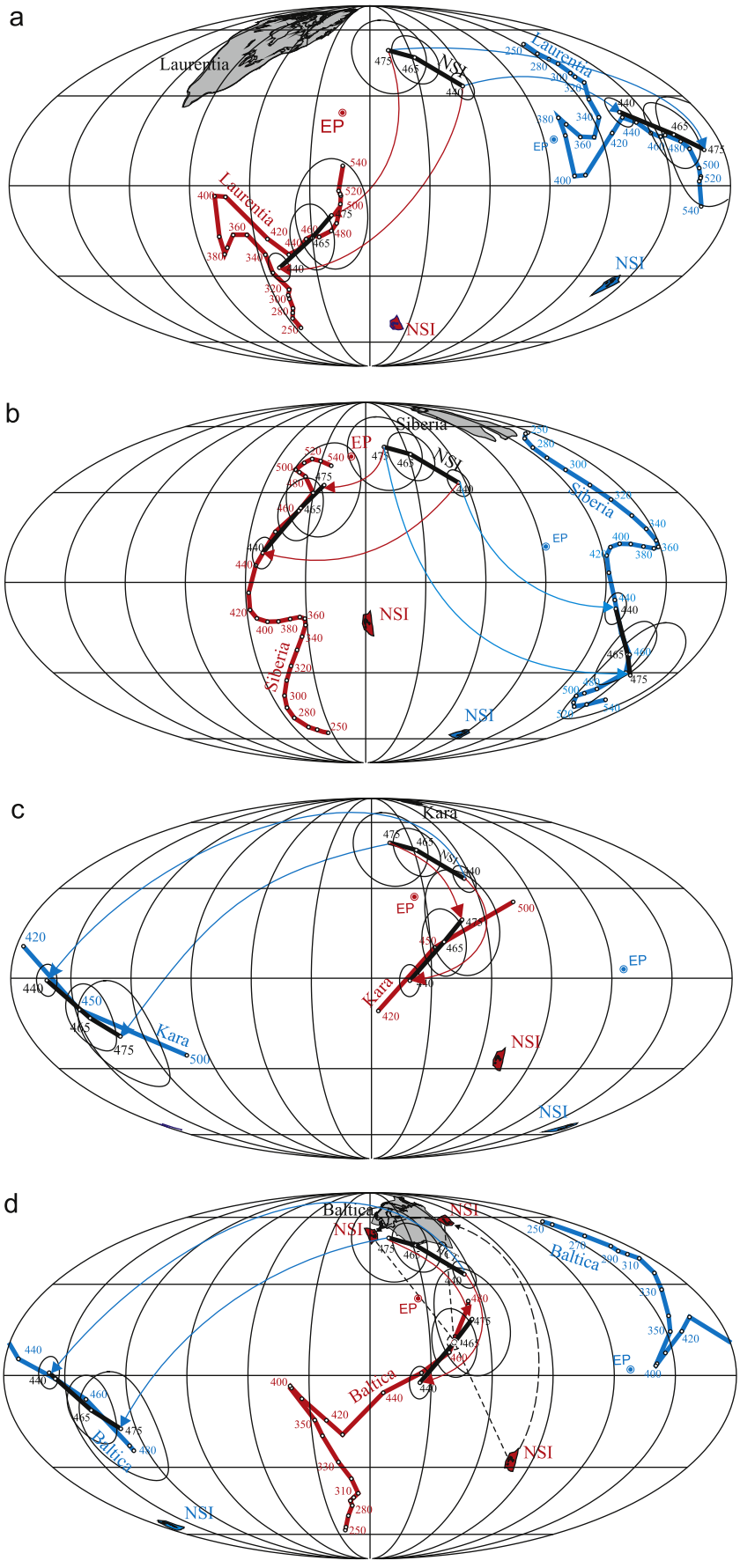
The site 11ns07 (lower part of the section) is characterized by higher (~30 mT) MDFs (Fig. 7b). Only a high-coercive NRM component (ChRM) has been isolated. Such high coercivity is indicative for a SD (PSD) magnetite and probably hematite (Fig. 7b).

The fold test of McFadden (1990) on 23 samples is positive at the 99% level. The in situ statistics $\xi_1 = 10.784$ is greater than its critical value of 7.836, while the unfolded statistics $\xi_1 = 1.249$ is well below. The Fisher (1953) precision parameter k is 39.1 in stratigraphic coordinates and 22.6 in geographic coordinates. The fold test of Watson and Enkin (1993) based on 1000 trials gives an optimum data grouping at 84.8% of unfolding with lower and upper 95% confidence limits at 79.4% and 89.9% respectively. The direction–correction tilt test of Enkin (2003) is also positive (*DC Slope* $= 0.851 \pm 0.293$, which is close to 1).

We interpret the high-coercive ChRM of the Ayan dolomites as the primary DRM (or PDRM) acquired at c. 475 Ma (Table 1). The mean paleomagnetic pole for the c. 475 Ma Ayan formation calculated by averaging of virtual geomagnetic poles (VGPs) is located at 48.9°N, 13.8°E ($A_{95} = 18.1°$).

3.2. Bennett Island, Middle Ordovician turbidites

The NRM of turbidites varies from 0.1 to 0.8 mA/m and their magnetic susceptibility is ~10^{-4} SI units. Q-values vary from 0.05 to 0.5 (triangles in Fig. 6). This suggests a chemical remanent magnetization (CRM) (Nagata, 1965). We believe that this CRM is related to the in situ


a
Laurentia
EP
NSI
Laurentia
b
Siberia
EP
NSI
Siberia
c
Kara
EP
NSI
Kara
d
Baltica
NSI
EP
Baltica

formation of authigenic magnetic minerals (Butler, 1992). As the AF demagnetization was not effective due to the presence of high-coercivity magnetic phase, we mostly used the thermal demagnetization.

Nearly half of the total remanence in the samples has been destroyed by heating to 280 °C. The remaining part of the NRM decayed rapidly after heating to 320–360 °C (Fig. 8). These unblocking temperatures may indicate Fe sulfides (phyrrotite, greigite) as the main magnetic carrier. Alternatively such decay can be caused by the inversion of the maghemite — the product of low-temperature oxidation or weathering (Dunlop and Özdemir, 1997; page 58) to hematite. Heating to temperatures of ~440–480 °C caused a dramatic increase of the remanence intensity and a chaotic behavior of its direction (not shown in Fig. 8) due to a laboratory-induced formation of a new magnetic phase (likely magnetite). A single interpretable remanence component has been isolated between 180 and 400 °C (Fig. 8). This component is characterized by consistent directions (Table 1). However, the demagnetizations of four samples from site 11ns12 did not reveal stable end-points and in this case we used the great circles method (Halls, 1976).

The fold test of McFadden (1990) is positive at the 95% level. The in situ statistics $\xi_2 = 2.243$ is greater than its critical value of 2.076, while the unfolded statistics $\xi_2 = 0.777$ is smaller. This test indicates that the ChRM predates the folding of the strata. The calculated paleomagnetic pole for the studied c. 465 Ma turbidites (45.5°N, 31.9°E, $A_{95} = 11.0°$) is about 12° south-east of the c. 475 Ma Ayan pole (see Section 3.1). This difference is compatible with the estimated average tectonic plate velocity in the Early Paleozoic (e.g. Condie et al., 2015). This together with a positive fold test suggests that the ChRM age is close to the rock age of c. 465 Ma.

3.3. Kotelny Island, Upper Ordovician — Llandoverian, Terutekh and Urasin formations

The NRM intensities and magnetic susceptibility values of these rocks are within 1–10 mA/m and $2–200 \times 10^{-6}$ SI units correspondingly (Fig. 6). They are rather consistent within individual sampling sites, but vary significantly between sites and form three clusters (Fig. 6). The Terutekh limestones from site 11ns01 have the lowest average magnetic susceptibility of 4×10^{-6} SI units and a relatively high mean NRM of 1.8 mA/m. Their *Q*-ratios are close to 10 and distributed along the lines of equal *Qn*. The mean values of NRM and χ of Terutekh limestones in site 11ns02 are close to those of Urasin limestones from site 11ns03 — 7 mA/m and 5×10^{-5} SI units correspondingly; their *Q*-ratios are close to 5. Urasin limestones from site 11ns04 have slightly lower mean NRM of 1.7 mA/m, but their magnetic susceptibility is close to that of Terutekh limestones. Consequently their *Q* ratios are just below 1, which is characteristic for the detrital remanence (Nagata, 1965).

Despite such difference in magnetic properties, all studied samples show similar behavior during the AF-demagnetization (Fig. 9). The MDF varies between 20 and 30 mT. After removing the low-stability viscous component AF-demagnetization isolated a single, stable, unipolar remanence component directed moderately downward to the west (Fig. 9). This remanence has not been completely destroyed even at 140 mT, which suggests the presence of highly paleomagnetically stable magnetic carriers such as SD or PSD magnetite.

The fold test of McFadden (1990) is positive at the 95% level. In situ statistics $\xi_2 = 3.045$ is greater than its critical value of 2.335 while unfolded statistics $\xi_2 = 0.843$ is below the critical value. The fold test of Watson and Enkin (1993) is also positive — the best grouping of paleomagnetic directions is achieved at 101.1% ± 22.0% of untilting. Application of the direction–correction tilt test (Enkin, 2003) also gives a positive result (*DC Slope*: 1.029 ± 0.515). This implies a pre-folding age (older than 364–342 Ma; Filatova and Khain, 2007) of the ChRM.

The ChRM direction of the c. 440 Ma Terutekh and Urasin limestones differs from those of the c. 475 Ma Ayan dolomites by over 25° in declination (Table 1). Based on a high paleomagnetic stability of these rocks and on positive fold tests we consider that the mean paleomagnetic pole, calculated by averaging four VGPs (Table 1) of 33.7°N, 55.7°E ($A_{95} = 5.1°$) has an age of c. 440 Ma. This paleopole is located south-east of the Ayan and Keppen Bay paleopoles (Fig. 10).

4. Discussion and tectonic implications

Most of the existing tectonic models consider the NSI (or at least its northern part — the De Long Archipelago) together with the Alaska–Chukchi terrane as a fragment of Arctida which broke away from Laurentia in the Late Jurassic during the opening of the Amerasian basin (e.g. Grantz et al., 1998; Shipilov, 2004). This fragment is thought to be a part of Laurentia since the Early Paleozoic. New paleomagnetic poles from southern (Anjou Archipelago) and northern (De Long Archipelago) parts of the NSI help to establish the interrelation of these parts in the Early Paleozoic and to test the alternative hypotheses about shelf continuation of the South-Anyui suture (SA1–3 in Fig. 1). Additionally a comparison of the NSI paleopoles with the Laurentian APWP provides a new view on the NSI–Laurentia paleogeographic relationship.

Adequate plate tectonics interpretation of the paleomagnetic data requires their sufficient (usually $>10^5$ years) time averaging. The average sedimentation rate in shelf environment is roughly estimated at ~10 mm/10^3 years (Lisitsyn, 2009). Hence in most cases over 1 m of sedimentary rocks must be sampled to average the geomagnetic secular variations (SV) at the time of sedimentation. Studied successions in Kotelny Island are significantly thicker than that, so they most certainly have recorded the time-averaged geomagnetic field. The sampled section in Bennett Island is thinner, but no evidence of an extremely rapid sedimentation has been observed. Therefore, we believe that our paleomagnetic data faithfully represent the time-averaged geomagnetic field.

According to the Geomagnetic Polarity Time Scale (GPTS) a long reverse polarity interval characterizes most of the Early and Middle Ordovician (Gradstein et al., 2004). This interval has been recorded in several Siberian localities (Pavlov et al., 2008) and sometimes is referred to as the Moyero Reverse Superchron (Pavlov and Gallet, 2005). A short normal polarity chron is documented at the beginning of the Tremadocian, so the c. 475 Ma Ayan paleopole from Kotelny Island may correspond to this chron. The following Moyero Reverse Superchron ended at c. 465 Ma and was followed by the long period of frequent reversals with some bias towards the normal polarity. Consequently the c. 465 Ma Keppen Bay Formation (Bennett Island), and the c. 440 Ma Terutekh and Urasin formations (Kotelny Island) could have been magnetized either in normal or in reverse geomagnetic field, but with slight preference to the normal polarity.

Importantly, regardless of polarity the closeness of c. 475, 465 and 440 Ma NSI paleopoles and their time–space progression (Fig. 10a) suggests that there were no significant differential movements within this terrane during the Ordovician times. Fig. 10a clearly demonstrates that it is likely that the NSI in c. 475–440 Ma has been a rigid entity.

The tectonic unity of the NSI is also supported by new ages of detrital zircons from Devonian–Carboniferous sedimentary rocks (Ershova et al., 2014a) of Kotelny Island and Cambrian–Lower Carboniferous

Fig. 11. The NSI APWP rotated to APWPs of (a) Laurentia (after Torsvik et al., 1996) by 15°N 95°E–125°(CW) (northern polarity option) and by 24°N 345°E–153° (southern polarity option); (b) Siberia (after Cocks and Torsvik, 2007) by 12°N 92°E–170° (northern option) and by 45°N 350°E–160° (southern option); (c) Kara (after Metelkin et al., 2005) by 3°N 126°E–147° (northern option) and by 27°N 24°E–122° (southern option); (d) Baltica (after Torsvik and Cocks, 2005) by 2°N 128°E–145° (northern option) and by 25°N 26°E–119° (southern option); northern poles and APWPs are in blue, southern in red. EP — locations of Euler poles of NSI rotation (the same color scheme).

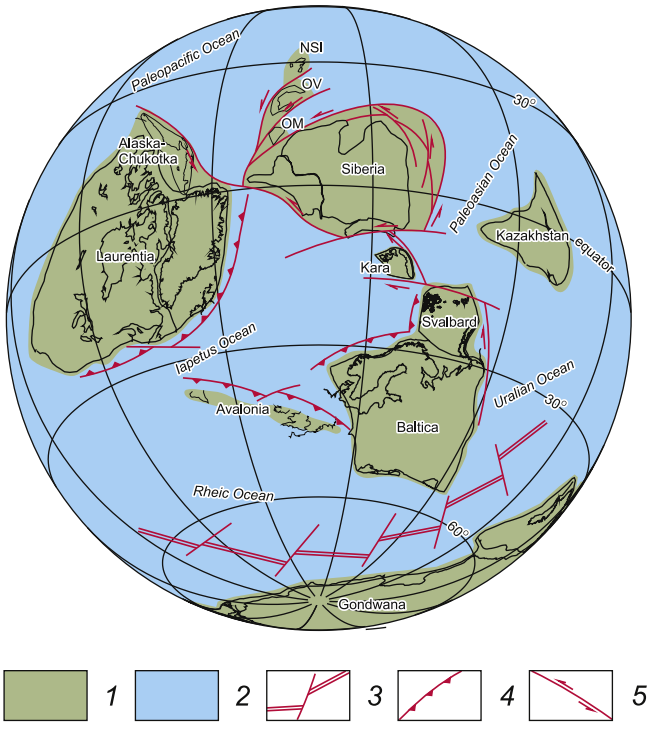


Fig. 12. Global paleogeographic reconstruction for the Late Ordovician (c. 450 Ma) after Metelkin et al. (2015). The NSI terrane is shown as a part of the Kolyma–Omolon superterrane including the Omolon (OM) terrane and Omulevka (OV) terranes (see Fig. 1). 1 — continents, 2 — oceanic basins, 3 — inferred position of spreading axis, 4 — active continental margins, and 5 — faults. The NSI terrane is rotated to the absolute framework about a pole 57.6°N, 30.1°E at 137.5° (CCW). Other Euler rotation parameters are as in Metelkin et al. (2015).

strata of the De Long Archipelago (Ershova et al., 2014b). These ages from both parts of the NSI are distributed between 1700 and 630 Ma and show identical peaks.

We used our paleomagnetic data to estimate the rates of the latitudinal drift (which represents a minimum plate velocity due to the absence of longitudinal motion constraints) and the angular velocity of the azimuthal rotation (i.e. rotation around the vertical axis located within the terrane) of the NSI terrane in Ordovician times. The results were <4 cm/year and ~2°/myr correspondingly (Fig. 10b). During the Early and Middle Ordovician, the azimuthal rotational component in the NSI motion prevailed, while later the latitudinal drift increased.

These calculations indicate that if we choose a normal polarity option, then in the Ordovician the NSI experienced a gradual, slow displacement in the intermediate latitudes with a relatively slow clockwise rotation (Fig. 10b).

To test the hypothesis about Laurentian affinity of the NSI we have to compare the newly available fragment of the NSI APWP (Fig. 10a) with the coeval segment of the Laurentian APWP. As the Ordovician NSI APWP is short and does not contain any peculiarities, within the error it can be compared not only to the Laurentian Ordovician APWP (Fig. 11a), but also to those of Siberia, Kara and Baltica (Fig. 11b–d). A still uncertain polarity of the NSI remanence makes the test even more difficult.

Regardless of the polarity choice, the APWPs' juxtaposition requires a significant spherical rotation of the NSI relative to the chosen craton (Fig. 11). Consequently, in both polarity options such juxtaposition implies a long distance between the NSI and Laurentia (Fig. 11a), or the other abovementioned continents (Fig. 11b–d). This can be most reasonably explained if the NSI terrane did not share the same lithospheric plate with any of those continents in the Ordovician and did not have common margins with any of them. However, if the NSI moved "independently", our paleomagnetic data allow its closer location to some of these continents due to the longitudinal uncertainty. In our case the NSI could be close to Baltica, if the magnetic remanence of the studied rocks has been acquired during the reverse polarity interval (Fig. 11d). If so, the NSI could have been located either near the south Scandinavian margin of Baltica, or near its Central Uralian margin (Fig. 11d). As Ershova et al. (2014a,b) suggested that if the detrital zircon found in the NSI Cambrian–Carboniferous deposits originated from Sveconorwegian, Timanian, and Caledonian orogenic belts, the likely

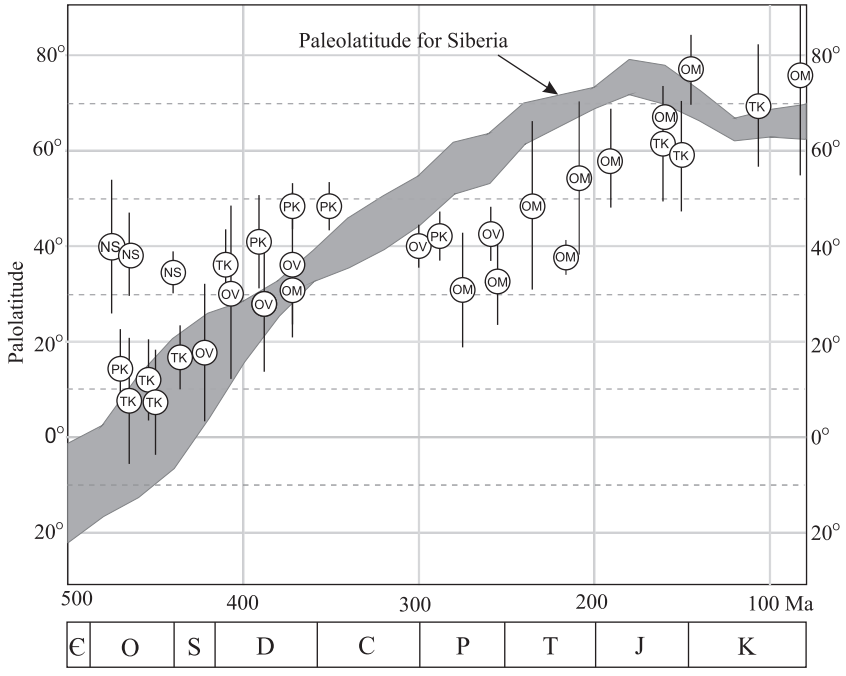


Fig. 13. Paleolatitudes (with 95% error bars, see Table 2) of Siberia (in the 64°N, 115°E reference point, gray band), NSI (NS) and terranes of the Kolyma–Omolon superterrane (abbreviations — as in Fig. 1).

Table 2
Early Paleozoic and Late Mesozoic paleomagnetic data from northeast Russia.

Target of research, age	Abb[a]	Sampling site		*N*	Paleomagnetic pole			Paleolatitude (°)	Reference
		φ (°N)	λ (°E)		*PLat* (°N)	*PLong* (°E)	*dp/dm* (°)		
New Siberian Islands									
Kotelny Isl., dolomite, Ayan fm., O_1	NS	76.2	139.0	43	48.9	13.8	18.4/13.7	39.9 ± 15.9	This paper
Bennett Isl. sandstone, Keppen bay fm., O_2	NS	76.2	149.1	23	45.5	31.9	13.6/9.9	38.0 ± 11.6	This paper
Kotelny Isl., limestone, Terutekh and Urasin fm., O_3–S_1	NS	75.6	137.9	27	33.7	55.7	5.5/3.8	34.4 ± 4.6	This paper
Kolyma–Omolon superterrane									
Prikolyma, sandstone, O_{1-2}	PK	65.1	152.2	19	34.2	13.1	6.5/12.0	14.3 ± 8.8	Rodionov and Shemyakin in Paleomagnetic directions and paleomagnetic poles (1989)
Tas–Khayakhtas, limestone, Tibitey fm., O_2	TK	67.5	139.4	6	−6	87	18.5/9.5	7.6 ± 13.2	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Tas–Khayakhtas, limestone, Yarlyk fm., O_3	TK	67.5	139.4	6	17.5	38.2	6.2/11.7	12.0 ± 8.5	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Tas–Khayakhtas, limestone, Middle Yarlyk fm., O_3	TK	67.3	139.2	3	11	354	15.4/7.9	7.6 ± 11	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Tas–Khayakhtas, limestone, Suruktyakh fm., S_1	TK	67.2	138.5	20	4	84	5.0/8.9	16.7 ± 6.7	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Omulevsky, limestone, sandstone, Mirnin fm., S_2	OV	63.5	149.5	7	−1.3	192.9	13.8/7.8	17.7 ± 12.2	Stone et al. (2003)
Tas–Khayakhtas, limestone, Datnin and Khobachalin fm., D_1	TK	67.2	138.5	27	23	90	12.9/9.3	36.5 ± 11	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Omulevsky, limestone, Neludim fm, D_1	OV	63.5	149.5	6	10	187.8	33.6/22.2	30 ± 25.4	Stone et al. (2003)
Prikolyma, basalts, Mokhovik fm. D_2	PK	64.4	151.2	20	38.5	225.0	19.4/14.7	41.0 ± 16.9	Kolesov (2005)
Omulevsky, basalt, Voyakh fm, D_2	OV	63.5	149.5	6	192.2	136	18.9/12.2	27.9 ± 15.2	Stone et al. (2003)
Omolon, andesite-basalts, Khelon fm, D_3	OM	65.0	162.0	44	20.6	224.0	10.4/7.9	30.3 ± 8.5	Kolesov and Stone (2002)
Prikolyma, andesite-basalts, Vorchun fm, D_3	PK	64.4	151.2	20	43.7	217.6	16.2/12.1	48.5 ± 12.7	Kolesov (2005)
Omulevsky, basalts, Chara fm, D_3	OV	63.6	149.3	6	11.7	170.8	26.1/18.6	35.9 ± 22.0	Stone et al. (2003)
Prikolyma, siltstones, sandstones, Duksundin fm, D_3	PK	64.4	151.2	27	50.9	232.0	11.3/7.2	48.0 ± 10.3	Kolesov (2005)
Omulevsky, basalt, Kip and Tur fm, C_3–P	OV	63.6	149.3	16	43	233.7	8.7/6.5	40 ± 7.5	Stone et al. (2003)
Prikolyma, basalts and tuffs, Chakhadan fm, C_3–P	PK	64.4	151.2	31	37.3	221.0	13.9/10.7	42.1 ± 5.7	Kolesov (2005)
Omolon terrane, siltstones, C–P	OM	65.5	156.8	8	40.9	262.3	14.7/9.8	30.8 ± 12.0	Shemyakin and Rodionov in Paleomagnetic directions and paleomagnetic poles (1989)
Omulevsky, argillite, limestones, Rogachev fm, P_2	OV	63.6	149.3	12	51.4	244.6	12.5/9.5	42.4 ± 10.9	Stone et al. (2003)
Omolon, siltstones, Khivach fm, P_2	OM	63.0	159.1	8	52.0	292.0	11.0/7.0	32.6 ± 9.1	Pechersky (1970a,b)
Omolon, siltstones and limestones, T_{1-3}	OM	63.0	159.0	12	56.0	248.0	19.6/16	48.3 ± 17.7	Pechersky (1970a,b)
Omolon, combined, T_3	OM	65.3	159.1	221	53.5	278.1	4.4/3.2	37.6 ± 3.7	Lozhkina, 1981
Omolon, siltstones and limestones, T_3	OM	63.0	159.0	22	58.0	235.0	17.2/14.8	53.9 ± 16.0	Pechersky (1970a,b)
Omolon, sandstones and siltstones, J_1	OM	65.0	155.4	36	40	122	10.7/9.6	58.6 ± 10.1	Didenko et al. (2002)
Tas–Khayakhtakh, combined, J_2	TK	67.3	139.5	33	46	106	12.8/11.7	61.8 ± 12.2	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Omolon, andesite-basalts and limestones, J_{2-3}	OM	63.0	159.5	12	64.0	103.0	3.7/3.5	66.7 ± 3.6	Pechersky (1970a,b)
Tas–Khayakhtakh, combined, J_2	TK	67.5	139.5	4	55.0	77.0	17.8/15.9	58.6 ± 16.9	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Omolon, sandstones and siltstones, J_3–K_1	OM	65.0	155.4	90	75	184	7.8/7.7	76.2 ± 7.7	Didenko et al. (2002)
Tas–Khayakhtakh, combined, K_1	TK	67.5	139.5	51	87.1	200.6	13.3/12.6	68.8 ± 13.0	Rodionov et al. in Paleomagnetic directions and paleomagnetic poles (1993)
Omolon, andesite-basalts and limestones K_2	OM	63.0	159.5	4	50	171	25.4/24.9	76.2 ± 21.3	Pechersky (1970a,b)

[a] Abbreviations are as in Fig. 1.

location of the NSI should be close to the northern margin of Baltica. This and other significant geological mismatches make unlikely both positions of the NSI shown in Fig. 11d. In particular, the abundance of endemic Siberian trilobite species from the Cambrian rocks of Bennett Island clearly indicates that the NSI was more likely adjacent to Siberia, than to Baltica (Danukalova et al., 2014).

According to most, if not all published Ordovician paleogeographic reconstructions a bulk of the continental crust has been located in the southern hemisphere (Cocks and Torsvik, 2002; Golonka et al., 2003; Lawver et al., 2002, 2011; Vernikovsky et al., 2013b; Metelkin et al., 2015). Consequently, if we suggest that the studied NSI rocks have been magnetized in the reverse geomagnetic field, the NSI should have been located either somewhere within the Iapetus Ocean between the active margins of Laurentia, Baltica and Avalonia, or within the Rheic or Ural paleooceans at a long distance from the northern margin of Siberia (modern Laptev Sea) (Fig. 12). In both cases the drift of the NSI terrane towards its present position with respect to Siberia would be too complicated.

Here we propose an alternative position of the NSI in the c. 450 Ma paleogeographic reconstruction of Metelkin et al. (2015) (Fig. 12). We suggest that the terrane has been located in the northern hemisphere. This reconstruction is paleomagnetically permissive, if the normal polarity option is suggested for our data. In our interpretation, the NSI terrane has been located within the Paleopacific near the southeastern (in modern coordinates) periphery of the Siberian paleocontinent. During the Ordovician and Silurian the NSI drifted towards Siberia with a simultaneous slow clockwise rotation (Fig. 11; see also Vernikovsky et al., 2013b; Metelkin et al., 2015). According to

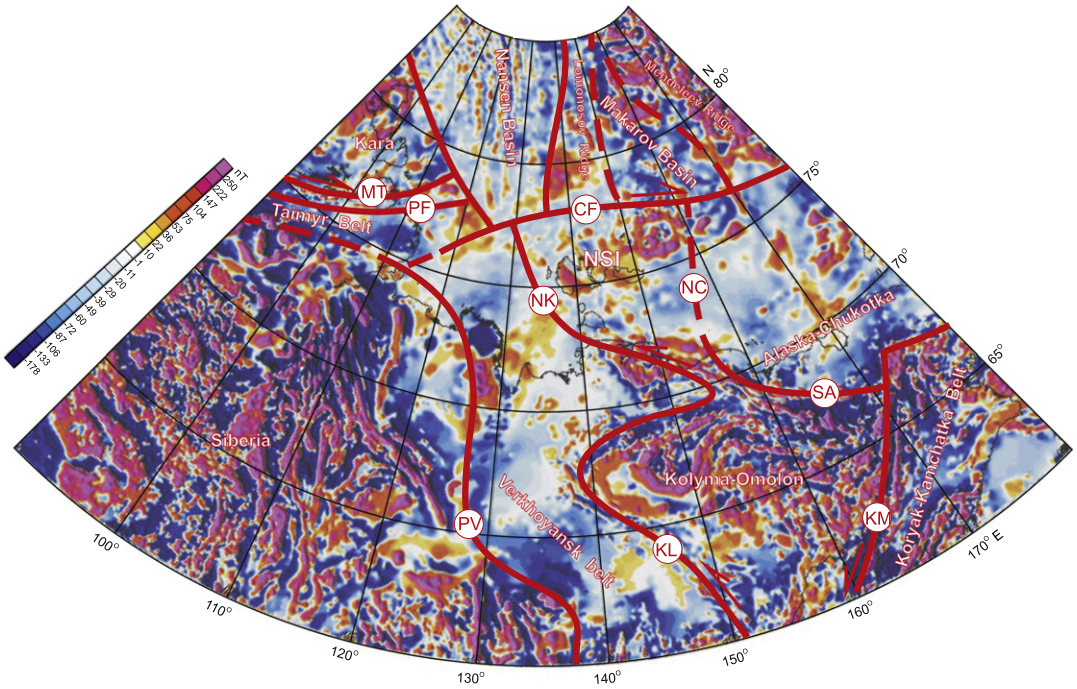


Fig. 14. A magnetic anomaly map (Gaina et al., 2011) with suggested margins of main tectonic units of northeast Russia including the Arctic shelf. CF — Charlie fault; KL — Kolyma Loop suture; KM — Koni–Murgal suture; MT — Main Taimyr suture; NC — New Siberian Islands–Chukchi suture (as the supposed extension of the South Anyui suture); NK — New Siberian Islands–Kolyma suture (as the supposed extension of the Kolyma Loop suture); PF — Pyasina–Faddey suture; PV — Priverkhoyansk thrust; SA — South Anyui suture.

published paleogeographic reconstructions the south-eastern parts of the Paleopacific in the Ordovician were characterized by relatively calm tectonic environments with prevalent passive continental margins on the Siberian shelves (Zonenshain et al., 1990). Several strike-slip systems could have been formed within the ocean basins around the Siberian craton (Fig. 12). Some small terranes like NSI could move closer to Siberia along these systems (Metelkin et al., 2012; Vernikovsky et al., 2013b; Metelkin et al., 2015).

The area bounded by the Kolyma Loop, South Anyui suture and Okhotsk–Chukchi volcanic belt is known as the Kolyma–Omolon superterrane (Fig. 1). The Precambrian Omolon microcontinent forms the core of this superterrane. The Prikolyma terrane and the dissected Omulevka terrane (Fig. 1) are both adjacent to the Omolon microcontinent. These terranes are composed mainly of Neoproterozoic and Paleozoic sedimentary rocks of a continental margin (Parfenov and Kuzmin, 2001; Oxman, 2003). The remaining part of the Kolyma–Omolon superterrane consists of island arcs' remnants and fragments of the oceanic crust. Several blocks (Omulevsky, Tas–Khayakhtakh, Selennyakh, and a few smaller ones) constitute the Omulevka terrane. These blocks are located along the Kolyma Loop (Fig. 1). Predominantly carbonaceous rocks of these blocks are similar to those in the NSI. The similarity of Devonian and older microfossils from the Siberian eastern (Verkhoyansk) continental margin with those from Omulevka and Prikolyma terranes suggests that these terranes were close to Siberia at least during the Early Paleozoic (Parfenov and Kuzmin, 2001). According to paleomagnetic data from the Late Silurian–Devonian rocks, these terranes could have drifted together coherently in moderate latitudes (Fig. 13) and could be close to the Siberian eastern margin (Stone et al., 2003).

There are only few Ordovician and Early Silurian paleomagnetic data from the north-eastern Asia (Fig. 1) and most of those are of a low reliability. However, some paleopoles from the Omulevka and Prikolyma terranes (Paleomagnetic…, 1989, 1993) are located close to and to the south of our NSI poles (Table 2). The difference in paleolatitude between these blocks and the NSI is just ~10° higher than their present latitudinal difference (~5°) (Table 2, Fig. 13). This is supportive to our hypothesis of the involvement of the NSI into the tectonic evolution of the Kolyma–Omolon superterrane. In particular this hypothesis suggests that in the Ordovician the Omulevka, Prikolyma and NSI terranes were separated from the Verkhoyansk margin of Siberia by a narrow ocean, which sometimes is referred to as the Oimyakon Ocean (Fig. 12). This suggestion is based on biogeographic similarities between the mentioned terranes (Danukalova et al., 2014).

Speculating further about paleogeographic affinity of the NSI and Omulevka terranes, we assume that the amalgamation of NSI with the Kolyma–Omolon superterrane took place in the Jurassic after the closure of the Oimyakon Ocean. Traces of this closure are manifested in the Uyandina–Yasachnen arc-related volcanic belt at the western (in present coordinates) margin of the Kolyma–Omolon superterrane. Magmatism in this belt is clearly suprasubductional and is related to the final stage of the formation of the Kolyma Loop (Polousny–Kolyma) suture (Parfenov and Kuzmin, 2001; Oxman, 2003). This collisional suture is marked by granitic plutons. The $^{40}Ar/^{39}Ar$ ages of these granites range from 143 to 138 Ma in the southern part of the Kolyma Loop, and from 138 to 120 Ma in the northern part (Layer et al., 2001). Such gradual decrease of age suggests an oblique collision, which implies a north-western (in present coordinates) drift of the Kolyma–Omolon superterrane. Traces of this drift are manifested in sinistral strike-slip faults found in the deformed structures of the Verkhoyansk–Kolyma orogen. Such sinistral Mesozoic shearing has been observed everywhere in the framing of the Siberian craton. These deformations can be explained by a clockwise rotation of the Siberian plate (Metelkin et al., 2010). The present day position of the NSI north of the Omulevka terrane could have resulted from the strike-slip displacements during and after the collision between Siberia and the Kolyma–Omolon terrane (Fig. 12). In these settings the trace of a tectonic suture between Svyatoi Nos Peninsula and Great Lyakhovsky Island (Fig. 1; Kuzmichev et al., 2005; Kuzmichev, 2009) should be a continuation of the Kolyma Loop rather than a continuation of the South Anuyi suture. The part of this line, located on the shelf is shown in Fig. 14 as the New Siberian Islands–Kolyma suture (NK). The continuation of the South-Anyui suture should be expected to the east of the

NSI. The exact shape and position of this continuation are ambiguous. We think that this suture follows close to and to the east of De Long Archipelago (SA1 in Fig. 1). We call it the New Siberian Islands–Chukchi suture (NC in Fig. 14).

The NSI terrane might have an extension on the northern outer shelf of the East Siberian Sea. This extension can include continental fragments in the southern part of the Lomonosov Ridge and, possibly in the Mendeleev Ridge. According to available geologic and geophysical data, these underwater ridges and the structures of the outer shelf are genetically related (Lebedeva-Ivanova et al., 2006; Poselov et al., 2012; Rekant and Gusev, 2012; Glebovsky et al., 2013; Vernikovsky et al., 2014). We also suggest that most of the Lomonosov Ridge is a fragment of the Kara continental margin that drifted away and was submerged to bathyal depths during the oceanic spreading in the Eurasian basin (Shipilov and Vernikovsky, 2010; Vernikovsky et al., 2013b; Sobolev et al., 2014). So, the northern margin of the NSI possibly lays somewhere north of the large recent Charlie Fault (CF in Fig. 14).

5. Conclusions

The New Siberian Islands and adjacent continental shelf are parts of a large terrane. A paleomagnetic study of Ordovician and Lower Silurian rocks in two distal parts of the NSI suggests a rigid integrity of this terrane. The Early Paleozoic sedimentary sequences exposed in Kotelny and Bennett islands have been deposited in the same marine basin. The NSI terrane has been located at ~30–45° N since c. 475 Ma until c. 440 Ma. Paleomagnetic data indicate that the NSI could not be part of Laurentia, Baltica, or Kara, but together with the Omulevka, Prikolyma, and probably Omolon terranes were involved into tectonic evolution of the Kolyma–Omolon superterrane near the Siberian craton. This conclusion is supported by biogeographic data. The present day position of the NSI farther north of these terranes possibly resulted from the strike-slip displacements during or after the collision of the Kolyma–Omolon superterrane with the Verkhoyansk margin of Siberia in the Jurassic. The South Anyui suture does not cross the NSI terrane, but separates it from the Alaska–Chukchi terrane east of the De Long Archipelago. The known trace of a tectonic suture in the Laptev Sea south of the NSI is probably a continuation of the Kolyma Loop.

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